1. CUSTOMER SEGMENT

- INSURANCE COMPANIES
- INSURANCE CLAIM INVESTIGATORS
- FRAUD DETECTION TEAMS IN INSURANCE FIRMS
- FINANCIAL RISK MANAGEMENT DEPARTMENTS
- INSURANCE DATA ANALYSTS
- INSURANCE POLICY MANAGEMENT SYSTEMS

2. JOBS-TO-BE-DONE / PROBLEMS

- **Insurance companies receive large numbers of claims every day.**
- **Manually checking each claim for fraud is time-consuming and expensive.**
- **Fraudulent claims cause huge financial losses for insurance companies.**
- **Traditional rule-based systems cannot detect complex fraud patterns.**

3.TRIGGERS (TR)

- Increase in insurance fraud cases.
- Financial losses caused by fake claims.
- Need for automation in fraud detection.
- Growth of machine learning in financial systems.
- Requirement for faster claim verification systems.

4.EMOTIONS: BEFORE / AFTER

Before Using the System

- Uncertainty about claim authenticity
- High financial risk
- Slow claim investigation process
- Stress for fraud detection teams

After Using the System

- Faster fraud detection
- Increased confidence in decision making
- Reduced financial losses
- Improved operational efficiency.

5. AVAILABLE SOLUTIONS (AS)

- Manual claim verification by investigators
- Rule-based fraud detection systems
- Traditional statistical analysis
- Third-party fraud detection services
- Basic database checks and document verification

Limitations of these solutions

- Time-consuming process
- High operational cost
- Low accuracy for complex fraud patterns
- Requires large human effort
- Slow decision making

6.CUSTOMER CONSTRAINTS (CC)

- Limited budget for advanced fraud detection systems
- Lack of skilled data science professionals
- Large volume of insurance claim data
- Need for faster claim processing
- Integration difficulty with existing systems
- Data privacy and security concerns

7. BEHAVIOUR

- Investigators manually check claim documents
- Cross-checking customer history and claim details
- Using internal fraud detection tools
- Performing risk assessment for suspicious claims
- Investigating abnormal claim patterns

8. CHANNELS OF BEHAVIOUR (CH)

Online

- Insurance management systems
- Web-based claim processing platforms
- Fraud detection software tools
- Data analytics dashboards

Offline

- Insurance company offices
- Claim investigation departments
- Customer service centers
- Insurance agents and field investigators

9. PROBLEM ROOT CAUSE (RC)

- Lack of automated fraud detection systems
- Increasing number of fake insurance claims
- Human errors in manual verification
- Complex fraud patterns difficult to detect manually
- Large amount of insurance claim data

10.SOLUTION S/L

- Develop a Machine Learning based fraud detection system
- Use Random Forest algorithm to classify claims as fraud or non-fraud
- Apply SMOTE technique to balance the dataset
- Build a Flask web application for claim prediction
- Provide real-time fraud detection results